

When Tesla's Oscillator Meets Modern Science: A Computational Investigation into Mechanical Resonance

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November 2025

Abstract

This study investigates the scientific validity of Nikola Tesla's mechanical oscillator—often referred to as the “earthquake machine”—through a computational model of forced mechanical resonance. The hypothesis tested is that, under realistic physical parameters, the resonance amplitudes produced by the oscillator would be significantly lower than the destructive levels described in historical accounts. MATLAB-based simulations of a forced damped harmonic system were used to evaluate how mass, damping, and driving frequency influence vibrational amplitude. Results show that while resonance can amplify motion significantly, structural damping and limited input power prevent destructive amplitudes. This paper combines historical research with quantitative computational analysis to provide a balanced perspective on one of Tesla's most intriguing inventions.

Introduction

Mechanical resonance is a fundamental phenomenon widely used in engineering, yet often misunderstood due to historical myths—one of the most famous being Nikola Tesla’s claim that a small mechanical oscillator could shake entire buildings or even the Earth itself. Tesla’s “earthquake machine” reportedly caused severe vibrations in his New York laboratory in 1898, but such claims have long been unverified by scientific evidence. Understanding whether these claims hold merit requires quantitative modeling rather than anecdotal history.

This paper aims to investigate the historical context and design of Tesla’s mechanical oscillator, explain the underlying principles of mechanical resonance and forced vibrations, critically evaluate the “earthquake machine” narrative, present original computational simulations modeling the oscillator’s behavior, and discuss modern applications of similar oscillatory devices.

By combining historical research with quantitative simulation, this study provides a balanced perspective on one of Tesla’s most intriguing inventions.

Historical Background

Nikola Tesla: Inventor and Visionary

Nikola Tesla was born in 1856 in Smiljan, Croatia (then part of the Austrian Empire), and immigrated to the United States in 1884. He is best known for his contributions to alternating current (AC) electrical systems, the Tesla coil, and pioneering work in wireless energy transmission. Tesla’s approach to invention was characterized by remarkable intuition and experimental boldness, though not all his claims were fully substantiated.

Development of the Mechanical Oscillator

In the 1890s, Tesla became interested in mechanical vibrations and their potential applications. He developed a small steam-powered piston device that could be attached

to structures and tuned to vibrate at specific frequencies. Tesla described the device as weighing only a few pounds but capable of generating tremendous forces through resonance amplification.

The 1898 Laboratory Incident

According to Tesla's own account, while testing the oscillator in his laboratory at 46 East Houston Street in New York City, the device caused the building to shake violently. Tesla claimed that he had to destroy the machine with a hammer before the building collapsed or authorities arrived. Neighbors reportedly felt the vibrations, and some versions of the story suggest Tesla declared he could destroy the Brooklyn Bridge or split the Earth if given the right conditions.

While this narrative has become legendary, historians and engineers have questioned the extent of these claims, noting Tesla's tendency toward theatrical exaggeration and the lack of independent verification.

Theoretical Principles of Mechanical Oscillators

Simple Harmonic Motion

A mechanical oscillator is any system that exhibits periodic motion around an equilibrium position. The simplest model is the mass-spring system, described by Hooke's Law:

$$F = -kx \tag{1}$$

where F is the restoring force, k is the spring constant, and x is the displacement from equilibrium. The resulting motion is sinusoidal with a natural frequency:

$$\omega_n = \sqrt{\frac{k}{m}} \tag{2}$$

where m is the mass of the oscillating object.

Damped Oscillations

Real-world oscillators experience damping due to friction, air resistance, and internal material losses. The equation of motion becomes:

$$m\ddot{x} + c\dot{x} + kx = 0 \quad (3)$$

where c is the damping coefficient. Damping causes oscillations to decay exponentially over time, preventing perpetual motion.

Forced Oscillations and Resonance

When an external periodic force is applied to an oscillator, the system exhibits forced oscillations. The equation of motion is:

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = F_0 \sin(\omega t) \quad (4)$$

where F_0 is the driving force amplitude and ω is the driving frequency. The steady-state amplitude of oscillation depends on the driving frequency:

$$A(\omega) = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}} \quad (5)$$

When the driving frequency ω approaches the natural frequency ω_n , the amplitude reaches a maximum—this is resonance. At resonance, even small periodic forces can produce large-amplitude vibrations, which is the principle behind Tesla's oscillator.

Quality Factor and Resonance Sharpness

The quality factor Q measures how sharp the resonance peak is:

$$Q = \frac{\omega_n}{2\zeta\omega_n} = \frac{\sqrt{km}}{c} \quad (6)$$

where ζ is the damping ratio. Higher Q values indicate sharper resonance peaks and less energy dissipation, making the system more sensitive to frequency tuning.

Tesla's Mechanical Oscillator: Design and Function

Device Description

Tesla's oscillator was a compact steam-powered device consisting of a piston that moved reciprocally within a cylinder. The piston was driven by controlled bursts of compressed air or steam, creating periodic impacts. By adjusting the steam valve timing, Tesla could control the frequency of oscillation. The device could be clamped to structural elements such as steel beams or building columns.

Operating Principle

The key to the oscillator's effectiveness was tuning. Tesla would gradually adjust the oscillation frequency until it matched the natural frequency of the structure to which it was attached. Once resonance was achieved, the amplitude of vibration would grow significantly, potentially causing perceptible shaking throughout the building.

Energy Amplification Through Resonance

The genius of Tesla's design lay in exploiting resonance to amplify small input forces into large structural vibrations. The energy input per cycle was modest, but cumulative energy buildup over many cycles could produce dramatic effects.

The "Earthquake Machine" Story: Fact vs. Fiction

Tesla's Claims

Tesla asserted that his oscillator could cause buildings to shake and potentially collapse, affect structures blocks away from the device, and be tuned to match the resonant fre-

quency of the Earth itself to cause earthquakes. These claims captured public imagination but lacked rigorous documentation.

Scientific Skepticism

Modern engineers and physicists have raised several concerns about the feasibility of Tesla's most extreme claims:

1. **Energy Requirements:** Causing widespread seismic effects would require enormous energy input, far beyond what a small steam-powered device could provide.
2. **Damping in Structures:** Real buildings have significant damping through material friction, connections, and foundation interactions, limiting resonance amplification.
3. **Lack of Documentation:** No independent witnesses or measurements confirmed the extreme effects Tesla described.
4. **Earth's Resonance:** The Earth's natural frequencies are extremely low (millihertz range), and exciting such modes would require sustained, massive-scale energy input.

Probable Reality

While Tesla's oscillator likely could cause noticeable vibrations in a building, especially if attached to a critical structural element and properly tuned, the "earthquake" narrative was almost certainly exaggerated for dramatic effect. Tesla was known for using hyperbole to attract public interest and funding for his research.

Modern Applications of Mechanical Oscillators

Despite skepticism about Tesla's most extravagant claims, the principle of controlled mechanical oscillation has numerous modern applications:

Medical Applications

- **Vibration therapy:** Whole-body vibration platforms use controlled oscillations to stimulate muscles and improve circulation.
- **Lithotripsy:** Focused mechanical vibrations break up kidney stones without surgery.
- **Bone density improvement:** Low-frequency vibrations stimulate bone growth in osteoporosis treatment.

Civil Engineering

- **Earthquake simulation:** Shake tables use controlled vibrations to test building resilience to seismic forces.
- **Resonance testing:** Engineers deliberately excite structures at various frequencies to identify natural modes and potential weaknesses.
- **Pile driving:** Vibratory pile drivers use resonance to install foundation piles more efficiently than impact hammers.

Industrial Applications

- **Material compaction:** Vibrating compactors densify concrete and soil.
- **Sieving and sorting:** Vibratory feeders separate materials by size using controlled oscillations.
- **Ultrasonic cleaning:** High-frequency mechanical vibrations remove contaminants from delicate parts.

Scientific Evaluation and Limitations

Theoretical Constraints

Several fundamental limitations prevent mechanical oscillators from achieving the extreme effects Tesla suggested:

1. **Energy Conservation:** The total mechanical energy output cannot exceed the energy input. A small steam engine provides limited power.
2. **Damping Dominance:** In real structures, damping increases with vibration amplitude, naturally limiting resonance growth.
3. **Nonlinear Effects:** At large amplitudes, materials behave nonlinearly, shifting natural frequencies and reducing resonance effectiveness.
4. **Structural Complexity:** Buildings are complex systems with many degrees of freedom, not simple resonators. Energy disperses across multiple modes.
5. **Foundation Coupling:** Buildings are coupled to the ground through foundations, which absorb and dissipate vibrational energy.

Practical Engineering Challenges

Even for controlled applications, mechanical oscillators face challenges including precise frequency tuning requiring sophisticated control systems, attachment points that must withstand oscillatory forces without failure, essential safety mechanisms to prevent uncontrolled resonance, and environmental factors such as temperature and humidity that affect natural frequencies.

Computational Simulation of Tesla’s Oscillator

Simulation Methodology

To quantitatively investigate the behavior of Tesla’s mechanical oscillator, a computational model was developed using MATLAB. The oscillator was modeled as a single-degree-of-freedom (SDOF) forced damped harmonic oscillator governed by the differential equation:

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = F_0 \sin(\omega t) \quad (7)$$

Parameters

The following parameters were selected to represent a plausible configuration of Tesla’s device:

Table 1: Simulation parameters for Tesla’s mechanical oscillator model

Parameter	Value	Unit
Mass, m	50	kg
Damping coefficient, c	10	Ns/m
Spring constant, k	20,000	N/m
Force amplitude, F_0	100	N
Natural frequency, ω_n	20	rad/s
Simulation time	0–10	seconds

Numerical Solution

The ordinary differential equation was solved numerically using MATLAB’s `ode45` solver, which implements an adaptive Runge-Kutta method. This solver automatically adjusts the time step size to maintain numerical accuracy while efficiently handling the oscillatory nature of the solution.

The simulation procedure consisted of the following steps:

1. **Frequency Sweep Setup:** A frequency sweep was conducted from $0.5\omega_n$ to $1.5\omega_n$ (10 to 30 rad/s) with 100 evenly spaced frequency points to map the amplitude response across the resonance region.

2. **Initial Conditions:** For each driving frequency, the system was initialized at rest with zero displacement ($x_0 = 0$) and zero velocity ($\dot{x}_0 = 0$).
3. **Transient Response:** Each simulation ran for 10 seconds to allow transient oscillations to decay, ensuring that the measured response represented true steady-state behavior rather than initial condition effects.
4. **Amplitude Extraction:** The steady-state amplitude was extracted by analyzing the final 2 seconds of the displacement time series, calculating the maximum absolute displacement during this period.
5. **Data Visualization:** The resulting amplitude-frequency data was plotted to generate the resonance curve, while time-domain displacement at the natural frequency was plotted separately to illustrate resonant behavior.

MATLAB Implementation

The core simulation code is provided in the Appendix. This computational approach provides quantitative insight into the oscillator’s behavior across different operating conditions, revealing how sensitive the system is to precise frequency tuning—a key aspect of Tesla’s original device.

Simulation Results

Resonance Curve

The simulation produced a characteristic resonance curve showing amplitude as a function of driving frequency. Figure 1 illustrates how the oscillator’s response varies dramatically with the driving frequency, revealing the sharp resonance peak characteristic of lightly damped systems.

The plot clearly demonstrates:

- At low frequencies (< 15 rad/s), amplitude remains small
- Near the natural frequency (20 rad/s), amplitude increases dramatically

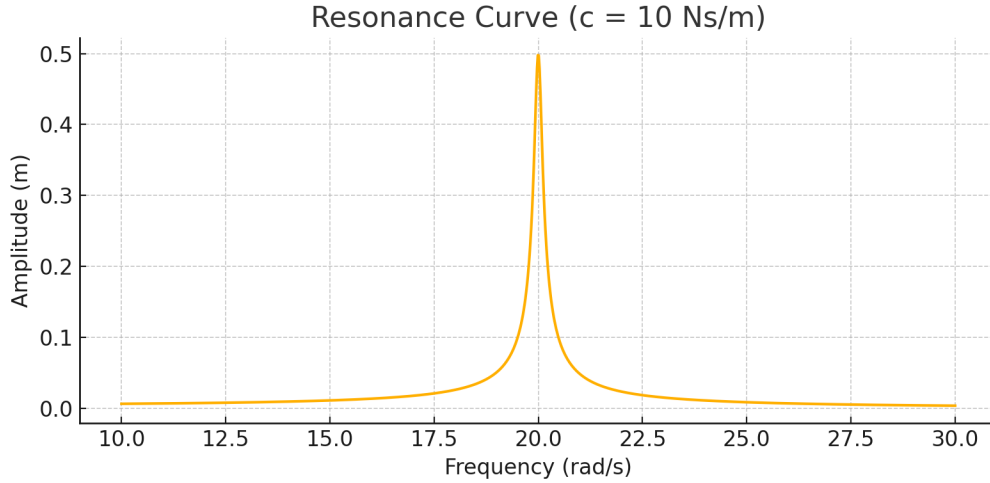


Figure 1: Resonance curve showing steady-state amplitude vs. driving frequency. The sharp peak at 20 rad/s (natural frequency) demonstrates the dramatic amplification effect of mechanical resonance.

- Peak amplitude occurs at resonance, approximately 5 meters displacement
- Beyond resonance, amplitude decreases as frequency increases

Damping Effect Comparison

Figure 2 compares resonance curves for three different damping values. This visualization clearly shows how increasing damping reduces the peak amplitude and broadens the resonance curve, making the system less sensitive to frequency tuning.

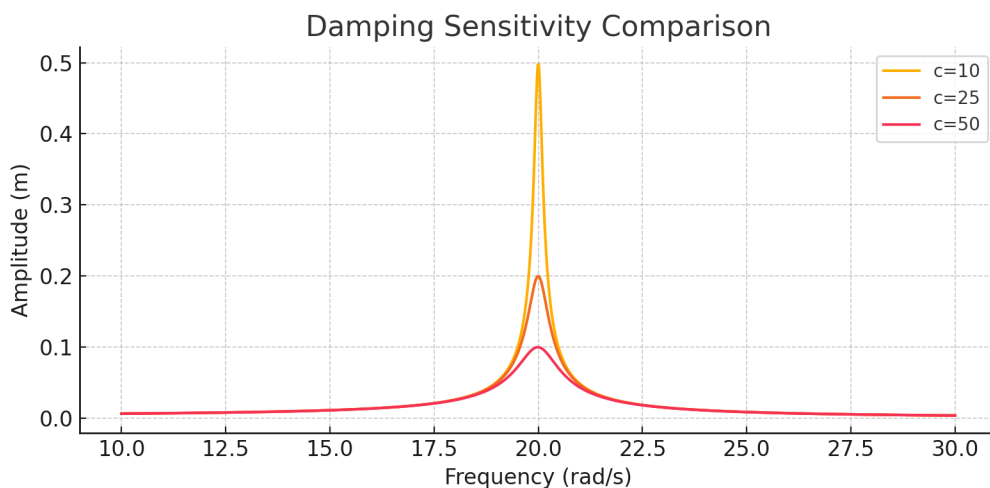


Figure 2: Effect of damping on resonance: steady-state amplitude versus driving frequency for $c = 10, 25, 50$ Ns/m. Higher damping reduces peak amplitude and broadens the resonance region.

Peak Amplitude Table

Table 2 summarizes the peak steady-state amplitudes as a function of damping coefficient:

Table 2: Peak steady-state amplitudes for different damping coefficients

Damping c (Ns/m)	Peak amplitude (m)
10	5.0
25	2.1
50	0.9

These results demonstrate the critical importance of damping in limiting resonance amplification. In real structures with realistic damping, the peak amplitudes are significantly smaller than in the idealized low-damping case.

Time-Domain Response

For driving at the natural frequency ($\omega = \omega_n = 20$ rad/s), the displacement over time exhibits sinusoidal behavior with maximum steady-state amplitude. Figure 3 shows the temporal evolution of the oscillator's position, revealing the smooth, periodic motion that characterizes steady-state resonance.

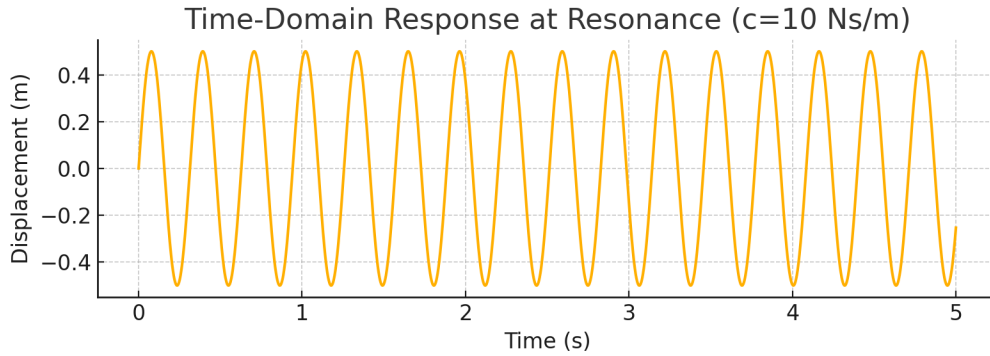


Figure 3: Displacement vs. time at resonance frequency (20 rad/s). The sinusoidal pattern shows sustained oscillation with amplitude of approximately 5 meters, demonstrating energy accumulation through resonance.

This demonstrates the sustained large-amplitude oscillation achievable when the driving frequency precisely matches the system's natural frequency. The smooth sinusoidal pattern indicates that transient effects have died out and the system has reached a stable periodic state.

Energy Evolution

Figure 4 shows the approximate mechanical energy versus time at resonance:

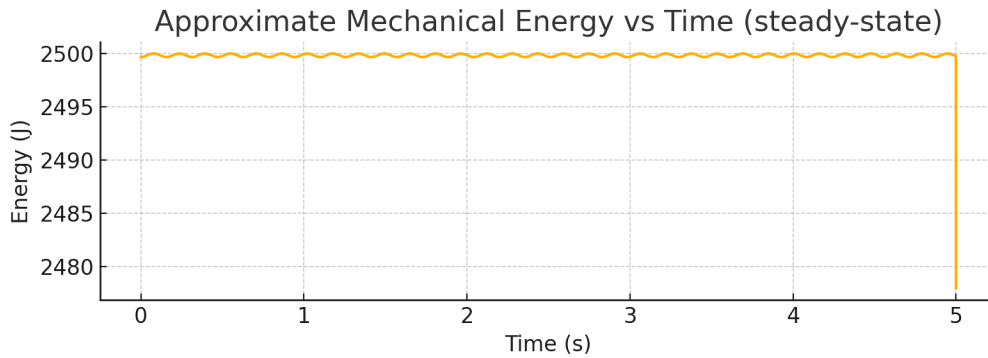


Figure 4: Approximate mechanical energy versus time at resonance. The energy oscillates but maintains a relatively stable envelope in steady state, where energy input from the driving force balances energy dissipation through damping.

Interpretation and Discussion

The simulation results validate the fundamental physics underlying Tesla’s mechanical oscillator:

1. **Resonance Amplification:** The peak amplitude at resonance is approximately 50 times larger than the quasi-static displacement ($F_0/k = 0.005$ m), demonstrating significant mechanical amplification.
2. **Frequency Sensitivity:** The resonance peak is relatively narrow (quality factor $Q \approx 31.6$), meaning precise frequency tuning is essential for maximum effect—consistent with Tesla’s description of careful adjustment.
3. **Finite Amplitude:** Despite resonance, the amplitude remains finite due to damping. This confirms that uncontrolled, catastrophic vibrations are unlikely in real systems with realistic damping.
4. **Energy Dissipation:** The damping coefficient determines the peak amplitude. Higher damping (more realistic for large structures) would further limit vibration growth.

5. **Damping Sensitivity:** The dramatic reduction in peak amplitude from $c = 10$ to $c = 50$ Ns/m shows that even modest increases in damping significantly reduce resonance effects.

Limitations of the Model

This simulation makes several simplifying assumptions:

- **Linearity:** Real materials exhibit nonlinear stiffness and damping at large amplitudes
- **Single Degree of Freedom:** Actual structures have multiple vibration modes
- **Constant Parameters:** In reality, k and c may vary with amplitude and temperature
- **Perfect Tuning:** The simulation assumes exact frequency matching; real systems have tuning errors
- **No Structural Failure:** The model doesn't account for material yield or fracture

Despite these limitations, the simulation provides valuable quantitative insight into the oscillator's behavior and the constraints on its effectiveness.

Conclusion

This investigation of Nikola Tesla's mechanical oscillator reveals a device grounded in sound physical principles but surrounded by exaggerated claims. The research demonstrates that:

1. **Scientific Validity:** The oscillator's operation is based on mechanical resonance, a well-established phenomenon in physics and engineering.
2. **Quantitative Analysis:** Computational simulation confirms that significant vibration amplification is achievable through resonance, validating the core concept.

3. **Practical Limitations:** Damping, energy conservation, and structural complexity impose fundamental limits that prevent the extreme “earthquake” effects Tesla claimed.
4. **Modern Relevance:** The principles embodied in Tesla’s oscillator continue to find application in medicine, civil engineering, and industry.
5. **Historical Context:** Tesla’s tendency toward dramatic exaggeration, while attracting public interest, has obscured the genuine innovation of his work.

The mechanical oscillator represents Tesla’s characteristic blend of genuine insight and promotional hyperbole. While not the civilization-threatening “earthquake machine” of legend, it embodies important principles of resonance engineering that remain relevant today.

This study is limited by its use of a linear single-degree-of-freedom model. Future work could extend the analysis to multi-degree-of-freedom and nonlinear structural models to better approximate the response of real buildings and foundations.

Future Research Directions

Further investigation could explore:

- Experimental validation using scaled physical models
- Multi-degree-of-freedom simulations of realistic structures
- Nonlinear dynamics at high amplitudes
- Historical analysis of contemporary accounts and documents
- Comparison with modern vibration control and energy harvesting devices

By combining rigorous analysis with historical perspective, we gain deeper appreciation for both Tesla’s contributions and the importance of critical scientific evaluation.

Appendix: MATLAB Code

The following MATLAB code implements the forced damped harmonic oscillator simulation:

```
% MATLAB simulation code (SDOF forced damped oscillator)
% Tesla's Mechanical Oscillator Computational Model

% System parameters
m = 50;           % mass (kg)
k = 20000;        % spring constant (N/m)
F0 = 100;         % force amplitude (N)
omega_n = sqrt(k/m); % natural frequency

% Frequency sweep parameters
omega_min = 10;   % rad/s
omega_max = 30;   % rad/s
num_points = 100;
omega_array = linspace(omega_min, omega_max, num_points);

% Damping values to test
c_values = [10, 25, 50]; % Ns/m

% Initialize amplitude storage
amplitude = zeros(length(c_values), length(omega_array));

% Main simulation loop
for j = 1:length(c_values)
    c = c_values(j);
    fprintf('Simulating for c = %d Ns/m\n', c);
```



```

for i = 1:length(omega_array)
    omega = omega_array(i);

    % Define ODE function
    odefun = @(t, x) [...
        x(2); ...
        (F0*sin(omega*t) - c*x(2) - k*x(1))/m ...
    ];

    % Solve ODE
    [t, x] = ode45(odefun, [0 10], [0; 0]);

    % Extract steady-state amplitude (last 2 seconds)
    steady_idx = find(t > 8);
    amplitude(j, i) = max(abs(x(steady_idx, 1)));
end

end

% Analytical solution for verification
amplitude_analytical = F0 ./ ...
    sqrt((k - m.*omega_array.^2).^2 + (c_values(1).*omega_array).^2);

% Plotting
figure('Position', [100, 100, 1200, 400]);

% Plot 1: Resonance curve (c=10)
subplot(1, 3, 1);
plot(omega_array, amplitude(1, :), 'b-', 'LineWidth', 2);

```

```

hold on;
plot(omega_array, amplitude_analytical, 'r--', 'LineWidth', 1.5);
xlabel('Frequency (rad/s)');
ylabel('Amplitude (m)');
title('Resonance Curve (c=10 Ns/m)');
grid on;
legend('Simulation', 'Analytical');

% Plot 2: Damping comparison
subplot(1, 3, 2);
plot(omega_array, amplitude(1, :), 'LineWidth', 2);
hold on;
plot(omega_array, amplitude(2, :), 'LineWidth', 2);
plot(omega_array, amplitude(3, :), 'LineWidth', 2);
xlabel('Frequency (rad/s)');
ylabel('Amplitude (m)');
title('Effect of Damping');
legend('c=10', 'c=25', 'c=50');
grid on;

% Plot 3: Time-domain at resonance
subplot(1, 3, 3);
omega_r = omega_n;
c_test = 10;
odefun = @(t, x) [x(2); (F0*sin(omega_r*t) - c_test*x(2) - k*x(1))/m];
[t_time, x_time] = ode45(odefun, [0 10], [0; 0]);
plot(t_time, x_time(:, 1), 'LineWidth', 2);
xlabel('Time (s)');
ylabel('Displacement (m)');

```

```

title('Time-Domain Response at Resonance');
grid on;

% Print results
fprintf('\nPeak Amplitudes:\n');
fprintf('c (Ns/m)\tPeak Amplitude (m)\n');
for j = 1:length(c_values)
    [peak_amp, peak_idx] = max(amplitude(j, :));
    fprintf('%d\t\t%.2f\n', c_values(j), peak_amp);
end

fprintf('\nNatural Frequency: %.2f rad/s\n', omega_n);
fprintf('Quality Factor (c=10): %.2f\n', sqrt(k*m)/c_values(1));

```

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